

Computer Network
* A collection of autonomous computers interconnected by a single technology.

* If two computers can exchange their information then they are interconnected.
* The aim is to share the resources among various devices.

Types based on size

- 1) PAN
- 2) LAN
- 3) MAN
- 4) WAN

1) PAN (Personal Area Network)

* Network arranged within an individual person.

* Range of 1-10 meters.

Types

- 1) Wired PAN (USB - Universal Serial Bus)
- 2) Wireless PAN (Wifi, Bluetooth)

↓
electromagnetic waves

* Wifi → Wireless Fidelity

2) LAN (Local Area Network)

* Group of computers connected to each other in a single building or office.

* Two or more PC's are communicated through a coaxial cable

* Inexpensive hardware such as hubs, network adapters & ethernet cables are used.

* Provides higher security.

* 10m - 10km

3) MAN (Metropolitan Area network)

* Covers larger geographical area by interconnecting different LANs to form a larger network.

* Used for communication ~~bet.~~ in the banks.

* Used in Airline reservation system.

* 1 - 10km

4) WAN (Wide area network).

* Covers large geographical area such as states or countries.

* Used in business, education & government

* 10 - 1000 km

Internet

* A global mesh of interconnected computer networks.

* LANs & WANs are interconnected to form a inter network (or) internet.

* 10000 km.

Uses of CN

→ Business Applications.

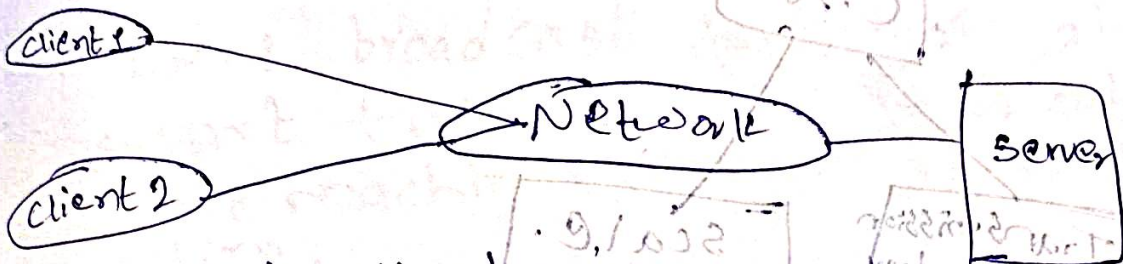
→ Home ~~issue~~ Applications.

→ Social issues.

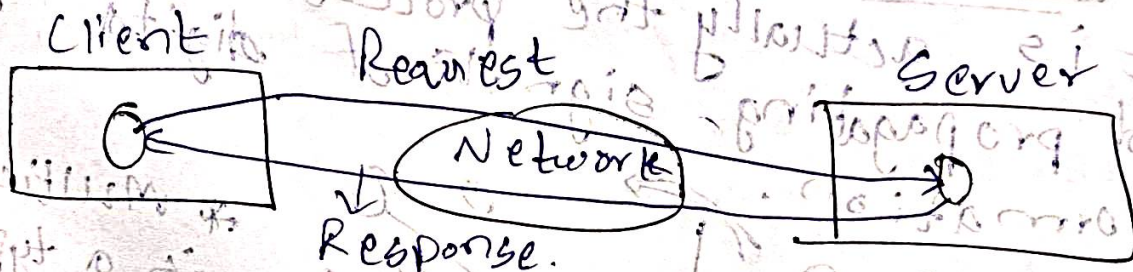
→ Mobile users.

Business Applications

- * Resources & data sharing.
- * A C.N can provide a powerful communication among employees.



- * Multiple clients can access the server with the same network.



Home Applications

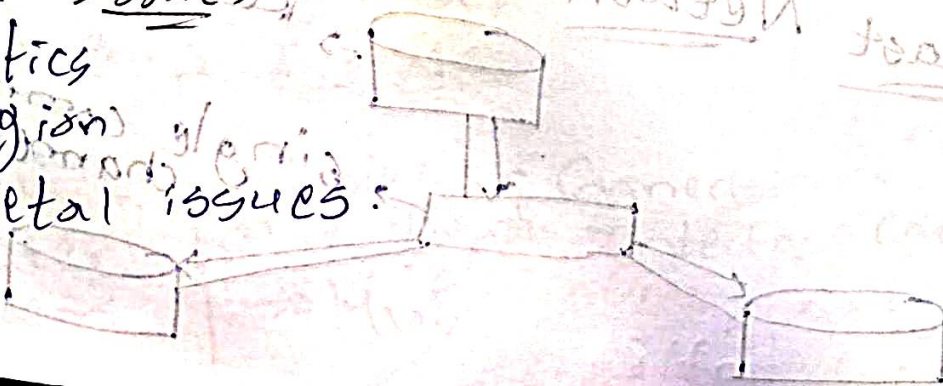
- * Access to remote info.
- * Person-to-person communication.
- * Interactive entertainment.
- * Electronic business.

Mobile users

- * Laptop, handheld computers.
- * Combination of wireless networks & mobile computing.

Social Issues

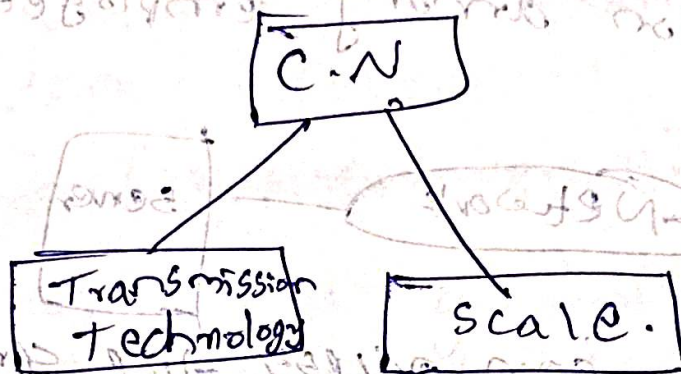
- Politics
- Religion
- Societal issues.



Network Hardware

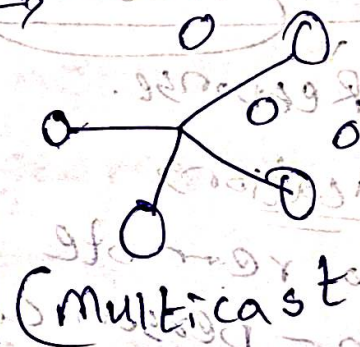
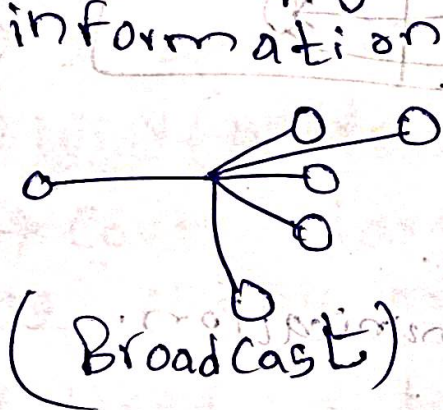
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* Deals with technical issues involved in network design.

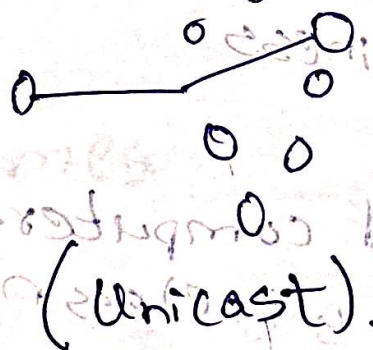


Transmission Technology

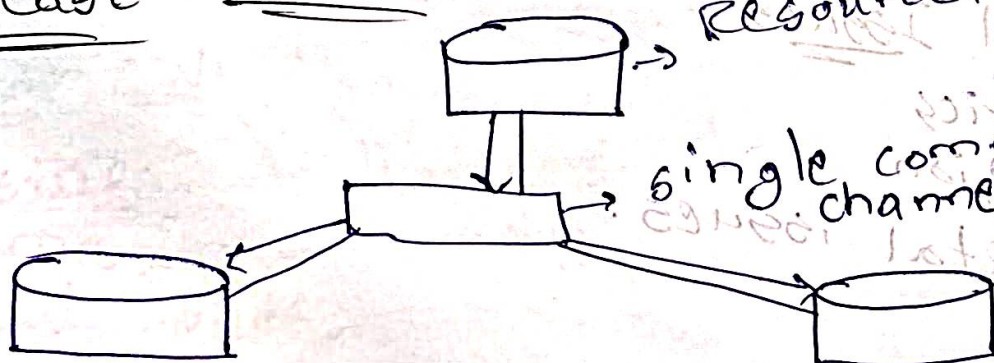
* It is actually the process of sending and propagating signals of digital information.



* Multicast is a type of broadcast.



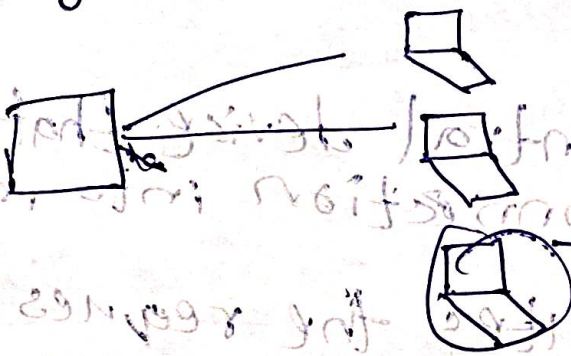
Broadcast Network



- * Packets (Short messages) sent by any machine is shared by all other resources or machines.

Multicast

- * Some broadcast systems also support transmission to a subset of the machines, known as multicasting.



→ This computer doesn't need the data or info so, it is not part of subgroup.

- * In broadcast even though the computer doesn't need the info it is connected to the server.

Point-to-Point Networks

- * Point-to-Point Networks consist of many connections between individual Pairs of machines.



(Unicast)

Components of C.N

- * Switch
- * Router
- * Modem

- * Cables & Connection
- * Network Interface Card
- * Hub

NIC (Network Interface Card)

- * It is a device that helps the computer to communicate with another device.
- * This contains hardware addresses.

Types

- 1.) Wireless
- 2.) Wired

Hub

- * This is a central device that splits the network connection into multiple devices.
- * Hub distributes the request to all the interconnected computers.

Switch

- * This is a networking device that groups all the devices over the network to transfer the data to another devices.
- * Directly sends message from source to destination.

Cables

- * Cable is a transmission media that transmits the communication signals.

Types of cables

- * unshielded twisted pair cable (UTP)
- * Coaxial cable.

* Fiber optic cable.

Mode

* Combination of modulation & demodulation.

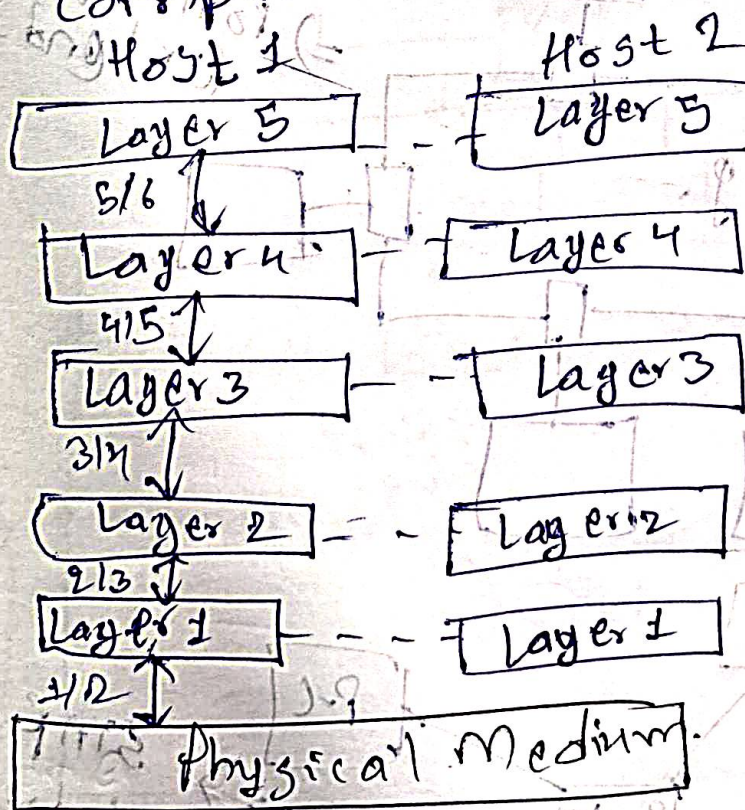
* This connects the computer to the internet over the existing telephone line.

Router

* This is a device that connects the LAN to the internet.

Network Software

* Software is used to control & manage all the resources used in a computer network.



Network Topology

* Topology: defines the structure of the network i.e; how all the components are interconnected to each other.

Types of Network

→ Bus

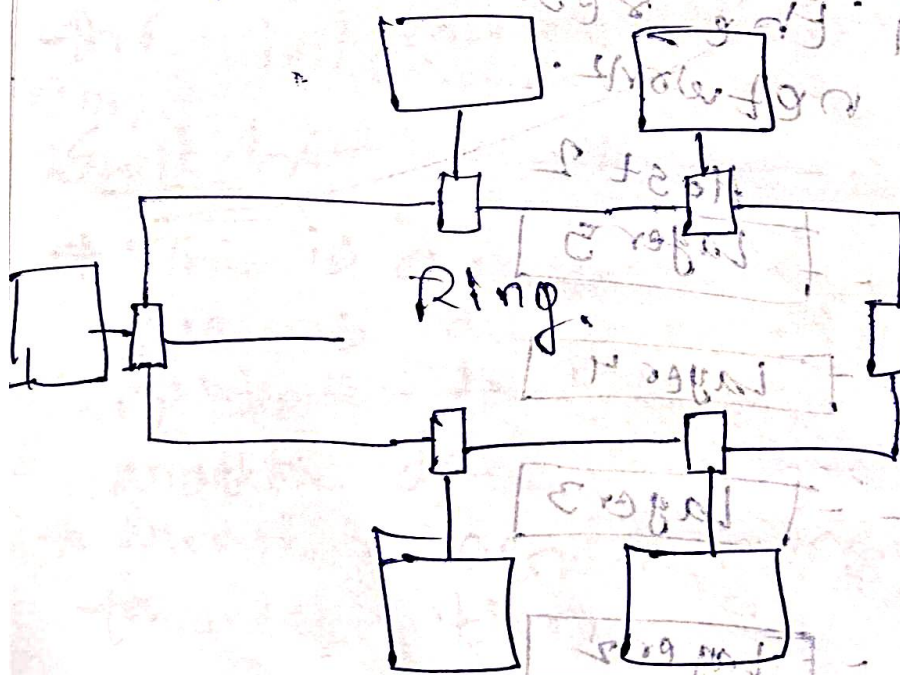
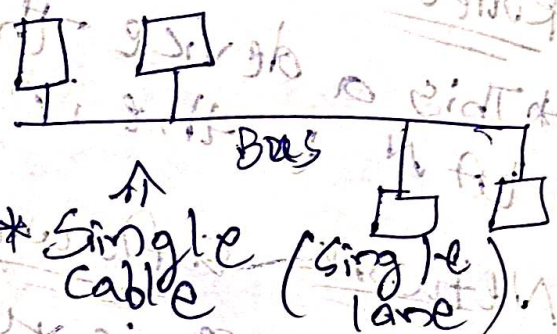
→ Ring

→ Tree

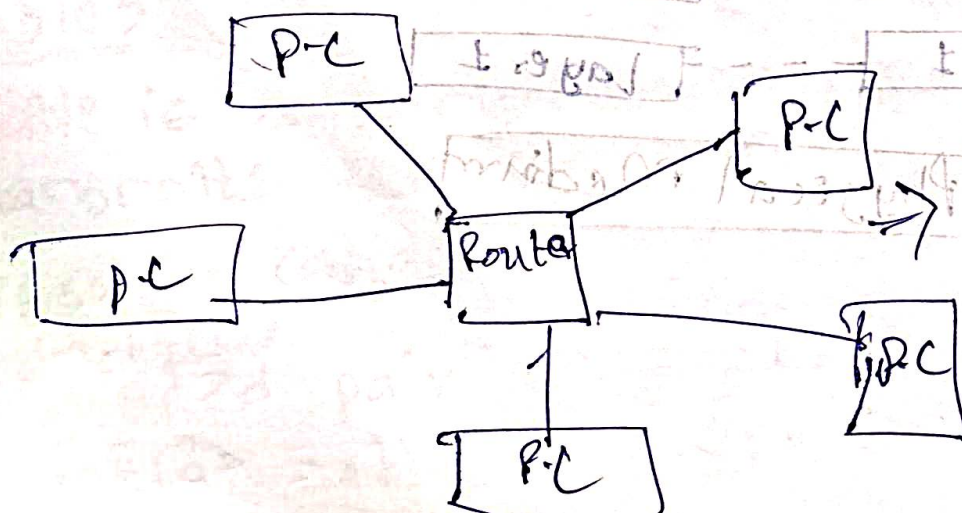
→ Star

→ Mesh

→ Hybrid

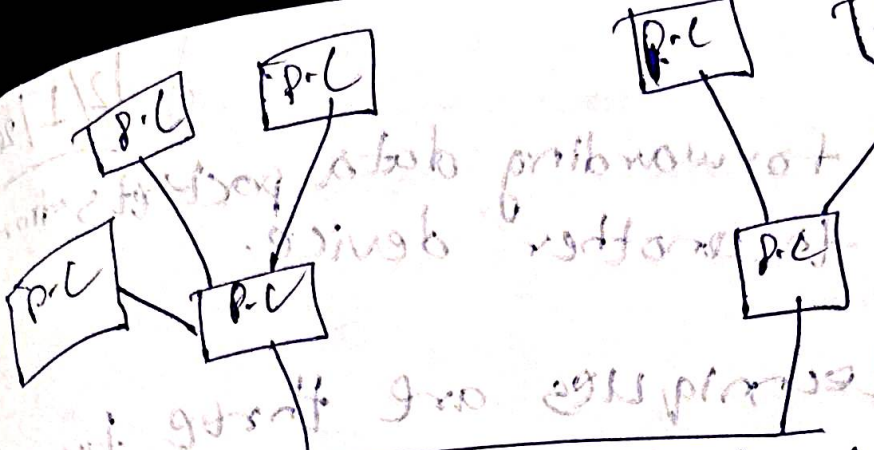


⇒ Point-to-point (or) end-to-end



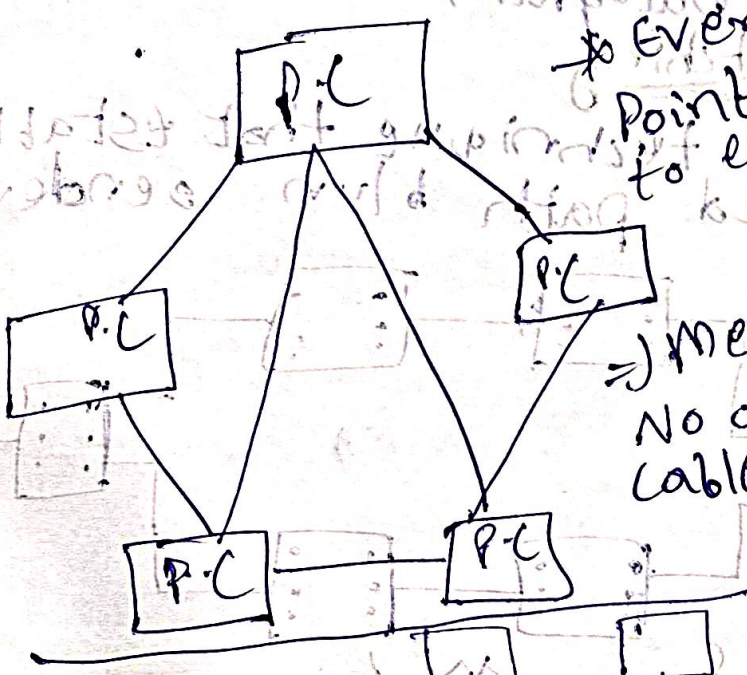
STAR.

* does not allow traffic blown devices.



* Combination of bus & star topology.

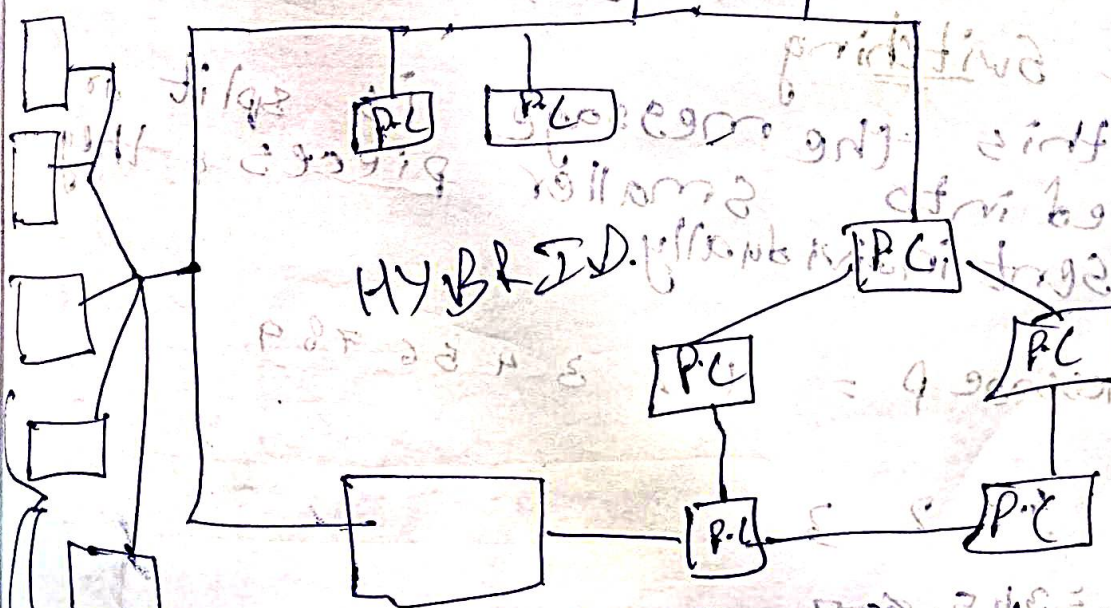
Tree Topology



Every device has a point-to-point link to every other device

= mesh

No of cables = $\frac{n * (n-1)}{2}$



HYBRID

* Combination of various topologies.

Switching

* Process of forwarding data packets from one device to another device.

* Switching techniques are three types.

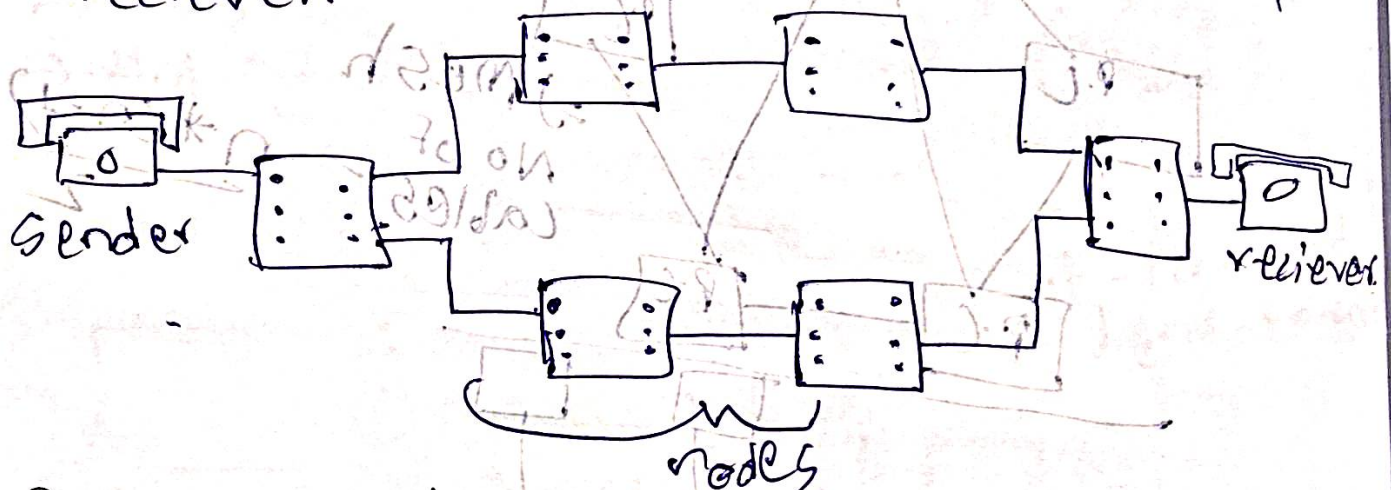
→ Circuit
→ message

→ Packet Virtual

Datagram

Circuit Switching

* This is a technique that establishes a dedicated path b/w sender & receiver.



Packet Switching

* In this the message is split or divided into smaller pieces & they are sent individually.

Given Message P = 1 2 3 4 5 6 7 8 9

F1 = 4 1 2 3

F2 = 3 4 5 6

F3 = 5 6 7 8 9

4 1 2 3 3 4 5 5 6 7 8 9 ⇒ Final byte sequence.

P = 5 6 7 2 5 1 3 5 6

F₁ = 3, F₂ = 2, F₃ = 4

F₁ = 4 5 6 7

F₂ = 3 2 5

F₃ = 5 1 3 5 6

4 5 6 7 3 2 5 5 1 3 5 6

* FLAG = @, ESC = \$

frame Format = FLAG DATA FLAG

D₁ = ABC ⇒ F₁ = @ABC@ ⇒ P₁ = ABC

D₂ = P@QR ⇒ F₂ = @P\$@QR@ ⇒ P@QR

* IF flag (@) is in between the data then we have to place Escape (\$)

D₃ = P@QR\$ ⇒ F₃ = @P\$@QR\$\$@

Transmission Modes

* The way in which data is transformed (or) transmitted from one device to another.

→ Simplex

→ Half-Duplex

→ Full-Duplex

Simplex

* In simplex mode the communication is unidirectional, i.e., the data flow is in one direction.

* Only data is sent not taken.

Half-Duplex

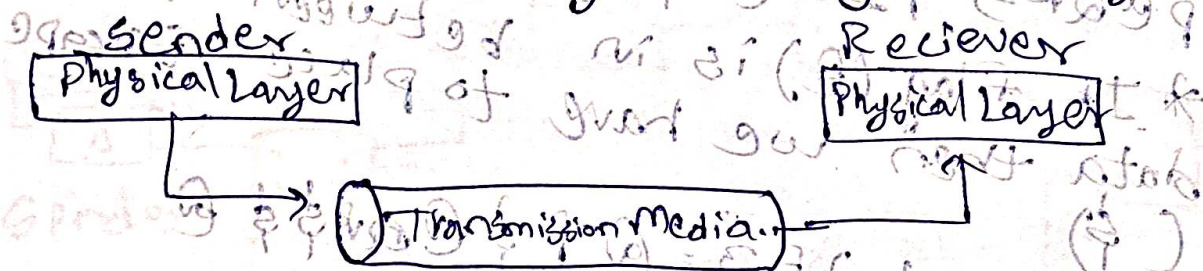
The station can transmit (and) receive the data as well. But not at the same time.

Full-Duplex

The communication is bi-directional, i.e., both stations can send & receive the message simultaneously.

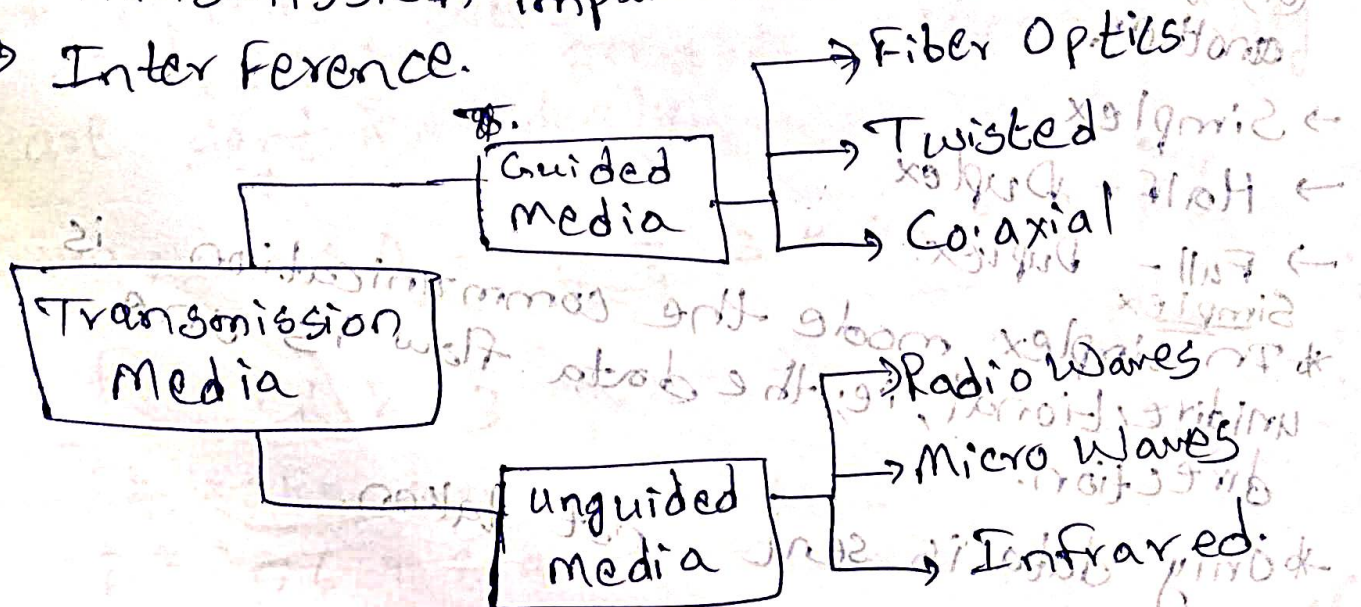
Transmission Media

* This is located by the physical layer & also controlled by the physical layer.



Factors

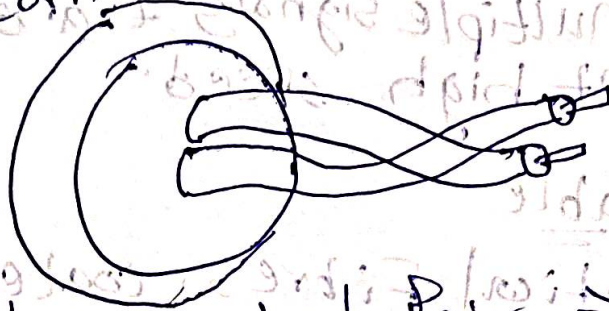
- Bandwidth ⇒ Refers to the maximum data transfer rate the media can handle.
- Transmission impairment
- Interference.



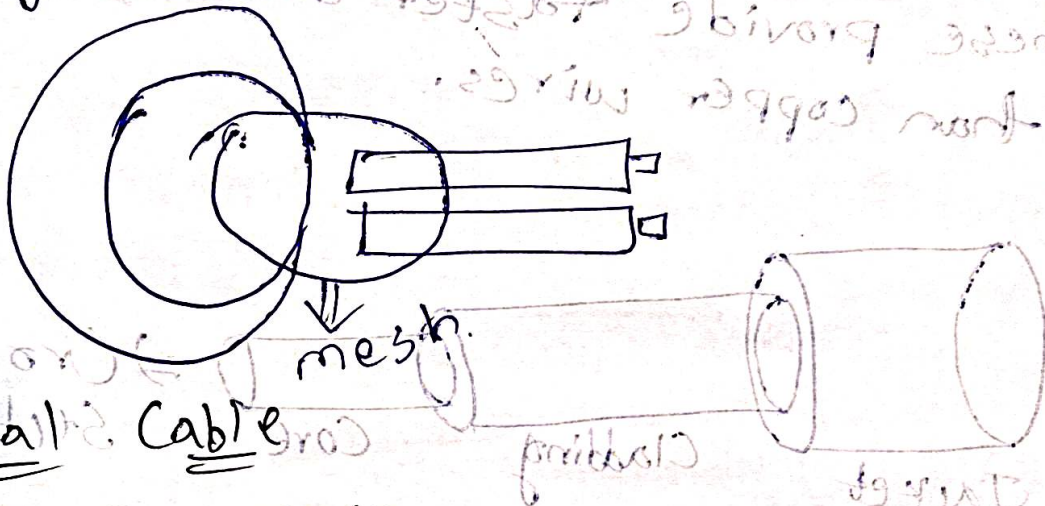
Guided Media

Twisted Pair

1) Unshielded Twisted Pair $\Rightarrow$ Used in telecommunication.

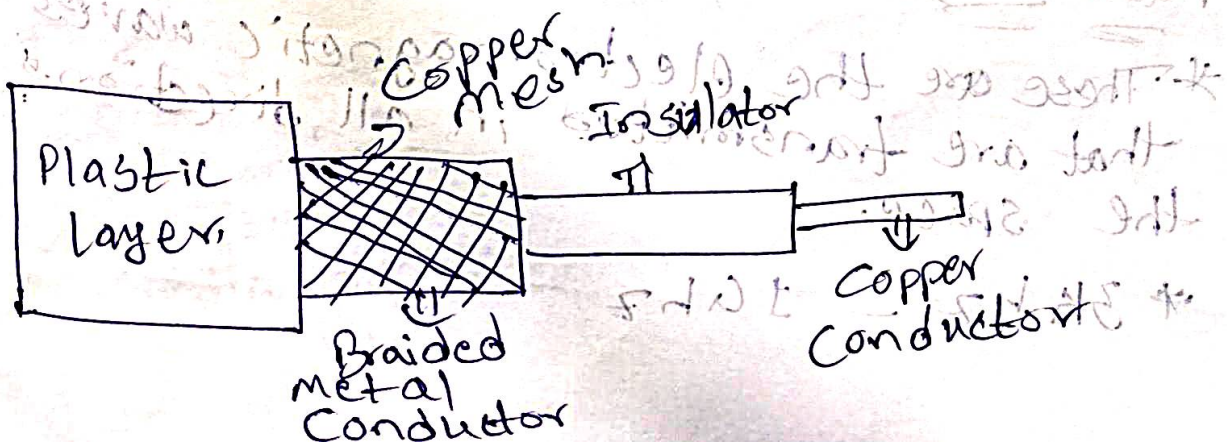


2) Shielded Twisted Pair $\Rightarrow$ Contains the mesh surrounding the wire that allows the higher transmission rate.



Coaxial Cable

- * Higher frequency
- * Contains two conductors parallel to each other.
- * Inner Conductor = Copper
- * Outer Conductor = Copper mesh
- * At middle insulator is placed.



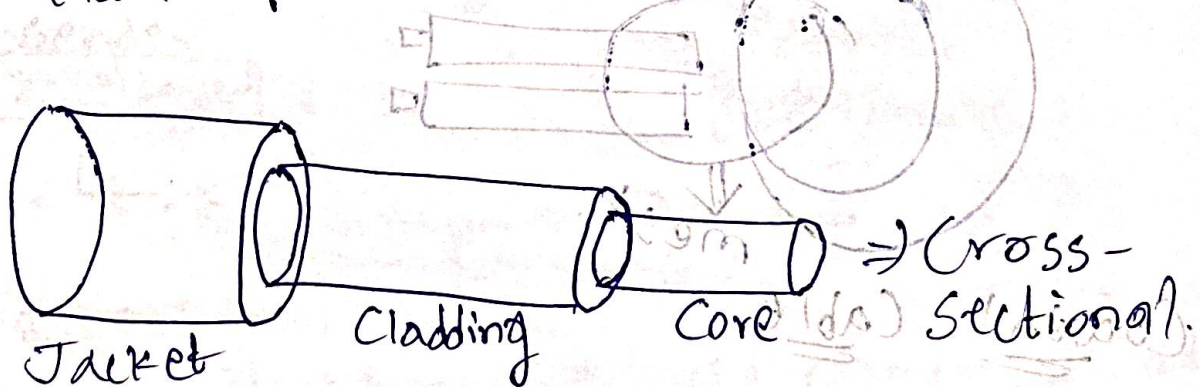
Types

1) Base band $\Rightarrow$ single signal transmission at high speed.

2) Broad band $\Rightarrow$ multiple signals transmission at high speed.

Fibre optic Cable

- * Holds the optical fibres coated in plastic that are used to send the data pulses of light.
- * These provide faster data transmission than copper wires.



- * Greater Bandwidth
- * Faster Speed
- * Better reliability.
- * Thinner & lighter.

Radio waves

- * These are the electromagnetic waves that are transmitted in all directions of the space.

* 3KHz — 1GHz



Micro Waves

* 1-300 GHz

* Microwaves are unidirectional

Infrared Waves

* 300 GHz - 400 THz

Line Coding schemes

Unipolar (-) All the signals levels are on one side of the time axis.

* NRZ (Non-Return-to-zero).

Polar $\Rightarrow$ Voltages are on both sides of the time axis.

* NRZ (Non-Return-to-zero).

* RZ (Return-to-zero).

Data link Layer

* This is responsible for reliable data transfer between two consecutive hops.

* Provides a well-defined service interface to the network layer.

* Services which are provided to network layer from data link layer are,

$\rightarrow$ Framing

$\rightarrow$ Addressing

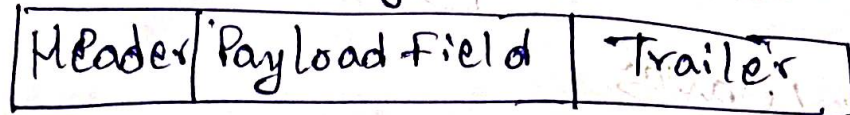
$\rightarrow$ Flow Control

$\rightarrow$ Error Control

$\rightarrow$ Media access control

Framing

* Breaking up the bit stream into portions is called Framing.



Methods

1) Byte count / Character count.

Message P = 1 2 3 4 5 6 7 8 9

Length of frames $\Rightarrow F_1 = 3, F_2 = 2, F_3 = 4$

$F_1 = (3+1)4$ | 1 2 3

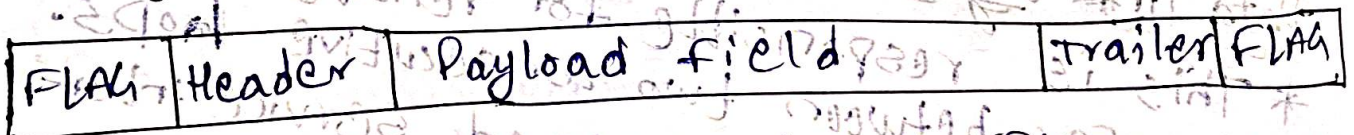
$F_2 = (2+1)3$ | 4 5

$F_3 = (4+1)5$ | 6 7 8 9

Final byte
sequence $\Rightarrow$

4 1 2 3 3 4 5 5 6 7 8 9

2) Byte stuffing / Character stuffing



Original
Bytes

After stuffing

[A] [FLAG] [B] $\rightarrow$ [A] [ESC] [FLAG] [B]

[A] [ESC] [B] $\rightarrow$ [A] [ESC] [ESC] [B]

[A] [ESC] [FLAG] [B] $\rightarrow$ [A] [ESC] [ESC] [ESC] [FLAG] [B]

[A] [ESC] [ESC] [B] $\rightarrow$ [A] [ESC] [ESC] [ESC] [ESC] [B]

Ex:- FLAG = @, ESC = \$, frame = FLAG DATA format = FLAG.

D1 = ABC | F1 = @ABC@

D2 = P@QR | F2 = @P\$@QR@

D3 = P@QR\$ | F3 = @P\$@QR\$\$@

Bit Stuffing

* In this framing we should ensure that flag pattern never appears inside the data.

Ex:- FLAG = 01111110

M1 = 1010100001110010111101

* As the sequence is not repeated no need to insert the zero to satisfy the rule as it already satisfies.

F1 = 01111110 1010100001110010111101 01111110

Ex:- 01111110 = FLAG

M2 = 10101111111011011111110000110

F2 =

01111110 10101111111011011111110000110 01111110

* Here the flag occurs so place a zero after 5 consecutive 1's as given in the flag to break the flag sequence occurrence.

Ex: FLAG = 011110
 M3 = 11001011110101110

F3) ~~01110 11001011101010101101~~

01110 1100101110101011100

011110

Examples

A: 01000111; B: 11100011; FLAG: 01111110;

ESC: 11100000;

* Four character = "A FLAG B ESC"
 on frame = 1 2 3 4

(a) Byte count

4+1 = 5 bytes

5 1111 A 1 0 FLAG B ESC

(b) Byte stuffing

FLAG A ESC FLAG B ESC ESC FLAG

(c) Bit stuffing

FLAG A FLAG B ESC FLAG

0111101 0100011 0111101 1110001 11100000 0111101

Error detection & Correction

* Data can be corrupted during transmission

Types of errors

* Single Bit error $\Rightarrow$ Only one bit in the data unit has changed

* Burst error $\Rightarrow$ Two or more consecutive bits in data unit have been changed.

Redundancy $\Rightarrow$ Adding extra bits for detecting errors at the destination.

* The important responsibilities of the data link layer are flow control & error control.

Flow Control $\Rightarrow$ Restricts the amount of data.

Error Control $\Rightarrow$ This is based on automatic repeat request, the re-transmission of data.

Data Link Layer Protocols

Protocols

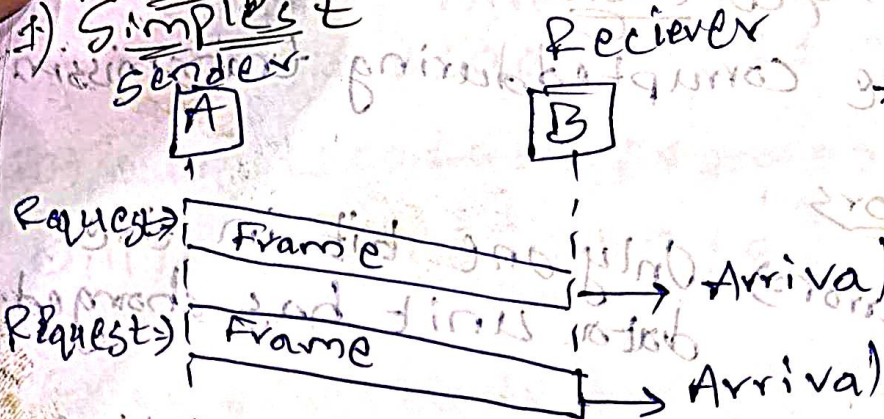
For noiseless channel

— simplest
— stop & wait

For noisy channel

— Stop & wait ARQ
— Go-Back-N ARQ
— selective Repeat ARQ

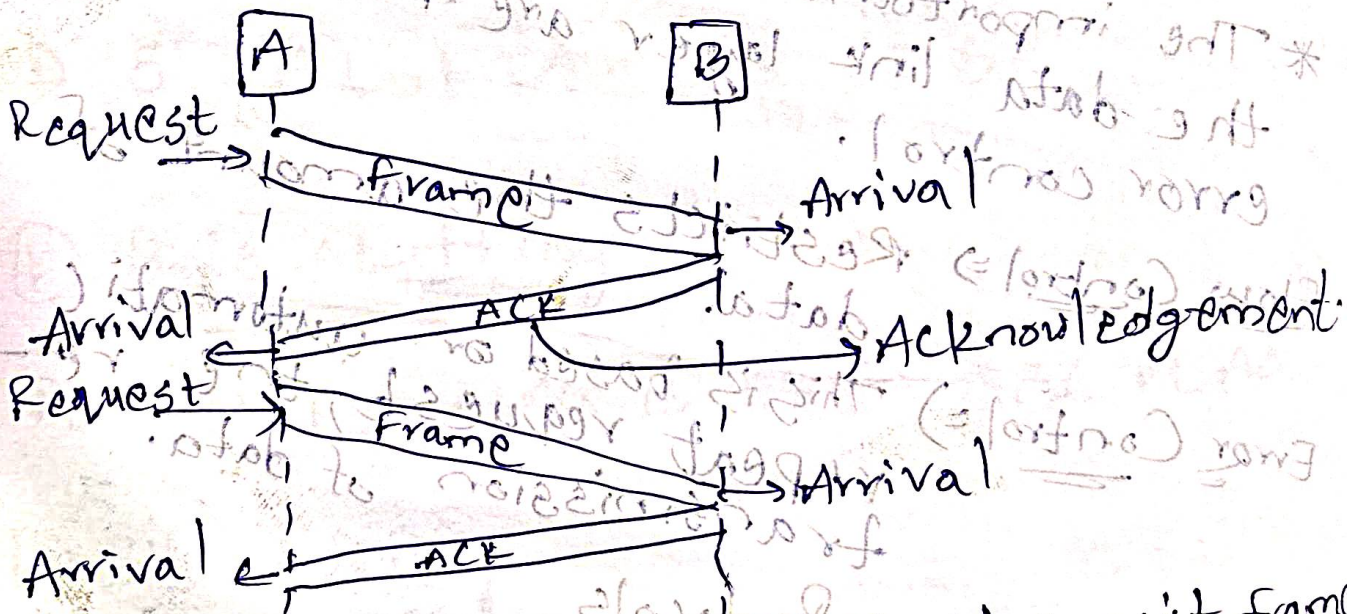
1) Noiseless & Simplex



* Very simple form of transmitting the data.

2) Stop-and-wait Protocol for noiseless channel

* Protocol with flow control.



* Sender will not transmit the next frame until it receives the acknowledgement of the last transmitted frame.

* Noiseless channels are non-existent.

Noisy

1) Stop-and-wait ARQ for noisy channels.

* Follows Automatic Repeat Request (ARQ).

* Same principle as stop-and-wait ARQ but that noiseless concept is non-existent.

Possibilities of the retransmission:

→ Lost frame (or) damaged.

→ ACK Lost.

→ Timer Time-out.

Sliding Window ⇒ This is a method of Flow Control in which a sender can transmit the several frames before getting an acknowledgement.

* A single ACK acknowledges multiple frames.

Sliding Window Protocols

* Used for continuous transmission error control.

1) Go-Back-N ARQ

* If one-frame is lost or damaged, then it retransmits all the frames starting from the damaged frame.

Possibilities ⇒ Damaged frame, Lost Data frame, Lost Acknowledgement.

2) Selective Repeat ARQ.

* This is more efficient than Go-Back ARQ.

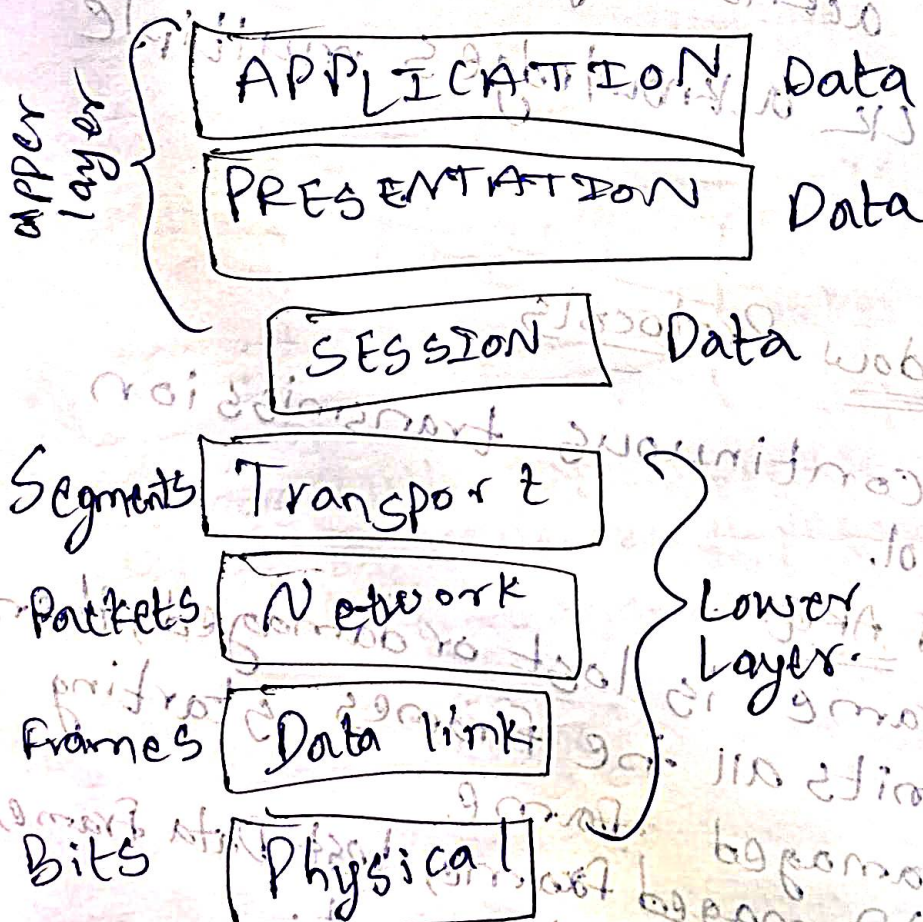
* In this, the frames only which have negative Acknowledgement (NAK).

* The receiver storage buffer keeps all the frames on hold until they are acknowledged.

OSI Reference Model

* This is established by the International Organization for standardisation (ISO)

* This covers all aspects of network communications. i.e. the open system interconnection (OSI) model.



Application Layer

- * Provides services to the user

Presentation Layer

- * Responsible for translation, compression, & encryption

Session Layer

- * Responsible for dialog control & synchronization

Transport Layer

- * This refers to end-to-end delivery of entire message

Network Layer

- * Responsible for the delivery of individual packets from the source host to the destination host

Data Link Layer

- * It is responsible for node-to-node delivery of data

Physical Layer

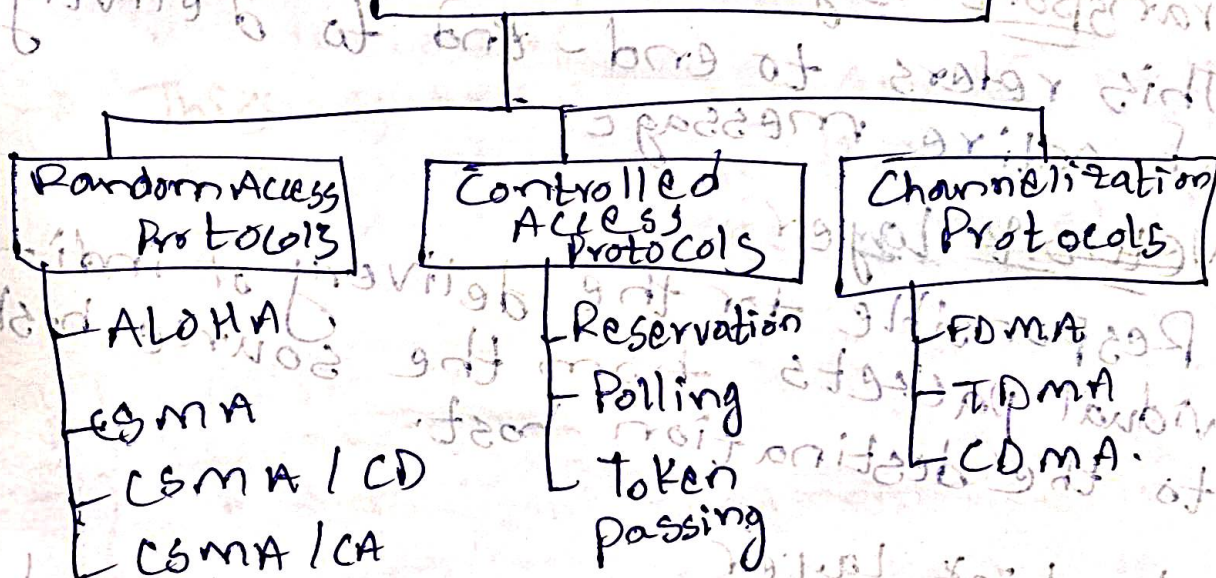
- * The physical layer is responsible for transmitting individual bits from one node to the next

* Data link layer is divided into two sublayers.

- (1) Logical Link Control Sublayer (LLC).
- (2) Medium access control sublayer (MAC).

* The function of MAC sub-layer is access control i.e., when multiple users/stations want to access the channel who should be allowed first & who should be allowed next.

Multiple Access Protocols



Random Access Protocols

* In this no station controls the channel, and no station permits or does not permit another station to use the channel.

* If more than one station tries to send, there is an access conflict, where ~~each station follows~~ the frames will be either destroyed or modified.

1) Pure ALOHA Protocol

* Pure ALOHA (original)

* Slotted ALOHA

→ Pure ALOHA is a simple protocol,
As per this protocol each station
sends a frame whenever it has to
send a frame.

→ In this only one channel is there
to share, so there is a possibility
of collision b/w frames.

→ If there is a collision
frames then the station waits for
a random amount of time & tries again.

Network Layer

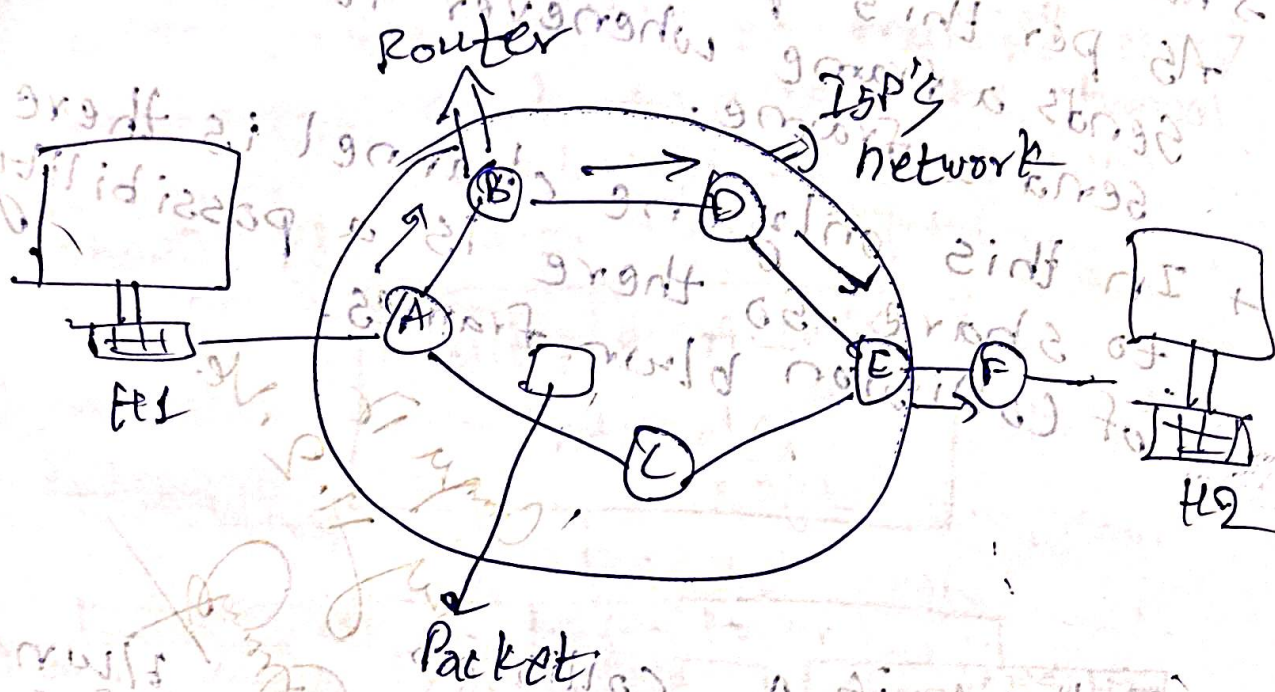
* This layer is concerned with getting
packets from the source all the way
to destination.

* Network layer must know about topology.

Design issues of network layer

1) Store-and-forward Packet Switching
If an Host 1 (a computer) wants to send
information to Host 2 (another computer),
that info or data is broken into
packets.

* There are multiple routers in the network (circle). The router receives the entire packet & stores it temporarily in memory & then it checks for errors in destination address, finally, it forwards it to the next router.



2) Services provided to the transport layer

* Services should be independent of the router technology.

* The transport layer should be shielded from the number, type, & topology of the routers present.

* While implementing the store & forward packet switching in a connection oriented service should follow the path & is fixed for every packet.

* Whereas, in connectionless service the path is not guaranteed or fixed path, packets can choose their own best paths.

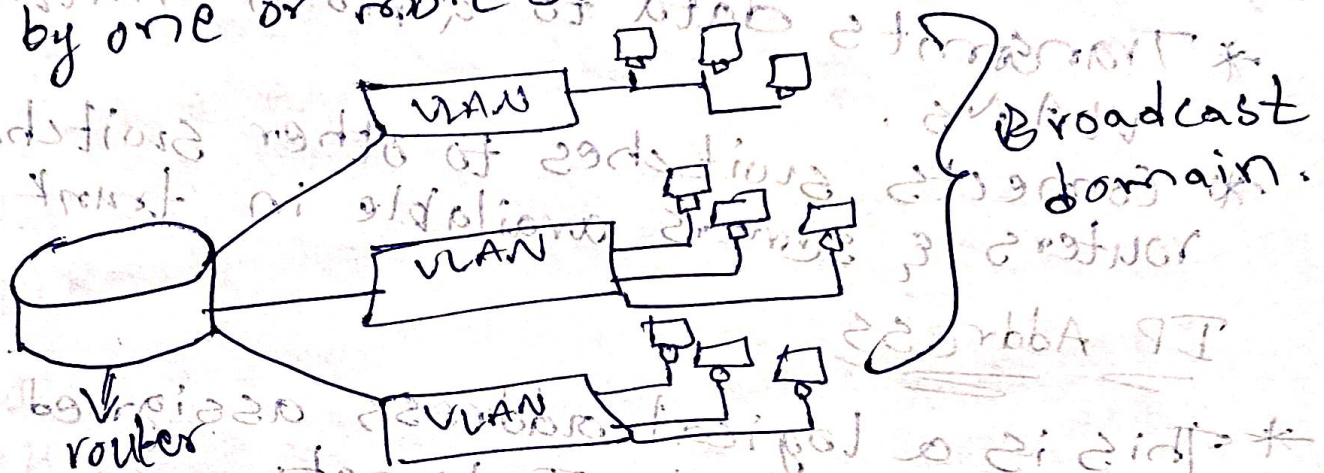
VLAN (Virtual Local Area Network).

* This divides one physical network into multiple logical networks.

* This is a logical segmentation of LAN.

* VLAN's address scalability, security & network management.

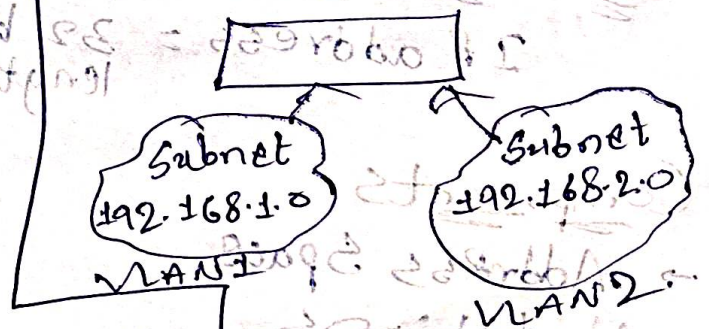
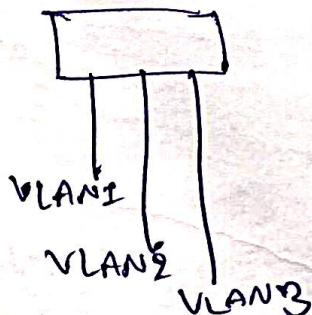
* A VLAN is a broadcast domain created by one or more switches.



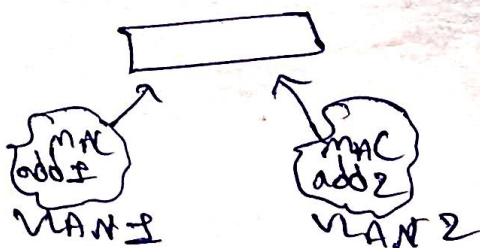
* Ports assigned to the same VLAN share broadcasts & vice versa.

Types

→ Port-based VLAN's → Protocol-based VLAN's



→ MAC address based VLAN's



Types of ports in VLAN

1) Access Ports

* This is a connection on a switch that transmits data to & from a specific VLAN.

* This connects switches to host devices.

2) Trunk Ports

* Transmits data to & from multiple VLANs.

* Connects switches to other switches, routers & servers available in trunk link.

IP Address

* This is a logical address assigned to every device connected to Internet.

* The length of IP address is dependent on the IP protocol.

Ex: In IPv4 protocol | In IPv6 protocol
IP address = 32 bits length | IP address = 128 bits length.

Components

- Address Space
- Notations
- Classful Addressing.
- Classless Addressing.

Address Space

* If we use N bits to define an address, then the no. of addresses in address space is 2^N .

IPv4 is 2^{32} .

Notations

* IPv4 address can be represented in 3 notations.

1) Dotted-Decimal

2) Binary

3) Hexadecimal

Dotted Decimal

* 4 bytes of an IPv4 address are separated by a decimal point or dot.

Ex:- 32.11.3.31

Binary

* Each byte is represented as an octet.

$$4 \times 8 = 32 \text{ bits}$$

10000000 00001011 00000011 00000001

198.11.3.1

Hexadecimal Notation

* Represented in 8 hexadecimal digits.

800B031F16.

Ex:-

Convert binary $\rightarrow$ Dotted-Decimal:

1) 10000001
 $255 \quad 64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1$

00001011
 $64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1$

0000101111101111
 $64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1 \quad 255 \quad 128 \quad 64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1$

129.11.41.239

Convert dotted-Decimal $\rightarrow$ Binary:

1) 111.56.45.78 $\rightarrow$ 10100110

01101111 00111000 00101101

IPv4 address rules

1) There must be no-leading zero(s).

Ex:- 111.56.045.78

2) There cannot be more than Four number (bytes).

Ex:- 221.34.7.8.20

3) Each number must be less than 255

Ex:- 75.45.301.14

4) A mixture of binary & dotted-decimal notations are not allowed.

Ex:- 11100010.23.14.67

Classful Addressing

* The address space is divided into five classes, A, B, C, D, E.

* Each class occupies some part of the address space.

→ If address in binary then first few bits will tell us the class of address.

→ If address is dotted-decimal, the first byte will tell the class.

	First byte	Second	Third	Fourth	First	Second	Third	Fourth
Class A	0				0-127			
Class B	10				128-191			
Class C	110				192-223			
Class D	1110				224-239			
Class E	1111				240-255			

(Binary) (Dotted-Decimal)

* In this IP address is divided into netid & hostid.

	netid	hostid
Class A	1	3
Class B	2	2
Class C	3	1

* Class D, E are not divided into netid & hostid.

D = Multicast.

E = Reserved.

Ex:-

1) 11000001 100000 11 00011011 11111111
Class C

2) 14.23.120.8

Class A

Range of Address

first Address = 146.102.129.0

Last Address = 146.102.32.255

Subtract

0.0.3.255

$$(0 \times 256^3 + 0 \times 256^2 + 3 \times 256^1 + 255 \times 256^0) + 1$$

$$= 1024$$

Ex.2:- 14.11.45.96 = first addresses

* number of addresses in the range is 32

$32 - 1 = 31$ as the last number should be less than maximum range ie; 32.

0.0.0.31

$14.11.45.96 + 0.0.0.31 = 14.11.45.127$

Bitwise Operations

1st no $\Rightarrow$ 00010001 01111001 00001110 00100011
 2nd no $\Rightarrow$ 11111111 11111111 10001100 00000000

Binary

1st no $\Rightarrow$ 17 121 14 35
 2nd no $\Rightarrow$ 255 255 140 0

17 121 12 0

128 64 32 16 8 4 2 0
 14 $\Rightarrow$ 0 0 0 0 0 0 0 0
 140 $\Rightarrow$ 128 0 0 0 0 0 0 0

Classless Addressing

* In this there are no ~~blocks~~ classes, but the addresses are still gained or granted in blocks.

* The addresses in a block must be continuous in nature.

* The no of addresses in a block must be a power of 2

* block can take any value from 0 to 32.

* /n defines prefix length

* /(n-32) denotes suffix length

* A block of address can be defined as

$X.y.z.b \Rightarrow$ defines one of the address
(n) $\Rightarrow$ prefix length (or) mask.

* No of address in the block can be found by using the formula:

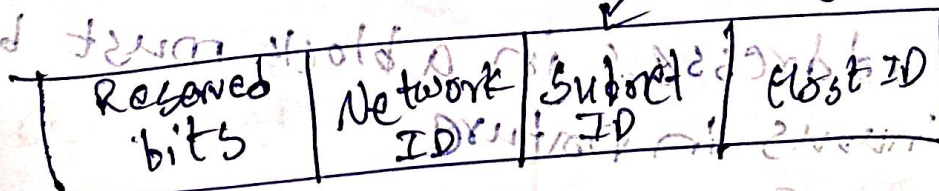
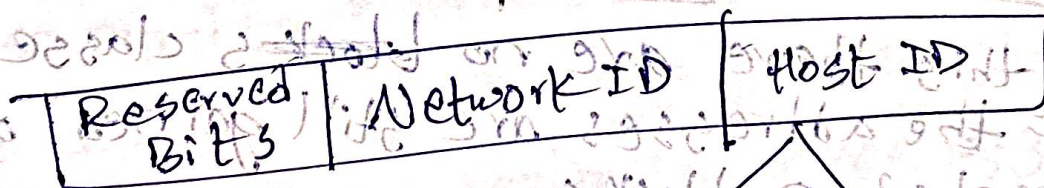
$$2^{32 - n}$$

Subnet:-

* Process of dividing a large network into smaller networks called subnets.

* Subnets provides each group of devices with their own space to communicate

* With subnetting, we split the network into four subnets, allocating just enough IP addresses for each department.



- NETWORK ADDRESS TRANSLATION
- * A process in which a local IP address is translated into a global (public) IP address.
 - * Translation of port numbers is also done.
 - * This generally operates on a router or firewall.

Port Address Translation

- * This is a feature of a network device that translates communications made between hosts on a private network & hosts on a public network.
- * NAT allows a single public IP address to be used by many hosts on the private network.

Address Mapping

- * The delivery of a packet to a host or a router requires two levels of addressing logical & physical.
- * This is done by either static (or) dynamic mapping.
- * Mapping Logical to Physical Address is done by address resolution Protocol (ARP).
- * Mapping Physical to logical address is done by reverse address resolution protocol (RARP).

* ARP request is broadcast
 * ARP reply is unicast
 * ARP table is maintained in the router
 * ARP table is updated on a regular basis

Routing
 * This is a process of forwarding data packets from source to destination
 * It involves the use of routing protocols
 * Routing is done by the router
 * Routing is done by the network layer

Routing
 * The delivery of a packet to a destination is done by the router
 * The router uses the routing table to find the best path
 * The router uses the routing table to find the best path
 * The router uses the routing table to find the best path

Routing
 * of network
 * for
 * incommen

Types
 * Non-
 * Adaptive

Non-
 * Route
 * on

Adaptive
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Routing
 1) Op
 * W
 * according

2) Flooding
 * Net
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